

ADNAN MASHRUR SADAD

MACHINE LEARNING ENGINEER | MODEL EVALUATION AND DATA SCIENCE

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Universiti Teknologi Malaysia (UTM), MJIIT

Kuala Lumpur, Malaysia

B.Sc. Computer Science (Software Engineering) with Honours — CGPA 3.43/4.00

2021 – 2026

- Dean's List 2022/2023 and 2025/2026. Thesis: OCR-based personal finance tracker performing automated, income-tax-ready expense classification.

TECHNICAL SKILLS

ML & Data: Machine Learning, Deep Learning, NLP, Python, SQL, scikit-learn, XGBoost, PyTorch, TensorFlow, pandas, NumPy, Hugging Face, computer vision, OCR

Evaluation: Temporal validation, baseline comparison, leakage analysis, drift detection (PSI/KS), experimental design, human annotation and inter-annotator agreement, LLM benchmarking

Engineering: FastAPI, Docker, Kubernetes, Kafka/Redpanda, PyFlink, Feast, MLflow, ONNX, REST APIs, CI/CD (GitHub Actions), Git, Agile

Other languages: Java, TypeScript, JavaScript, C++

RESEARCH

Deceptive Grounding in Regulatory RAG (SAFE-RAG)

2025 – 2026

First author — short paper under review at ALTA 2026, supervised at UTM MJIIT | [GitHub Link](#)

- Measured an 8.6% base rate (95% CI 4.5–13.5%) of answers that pass schema validation and are entailed by the passage they cite yet remain wrong, because that passage governs a different regulated entity — across 200 items annotated from 2,000 pipeline outputs over the ObliQA financial-regulation corpus.
- Ran the study pre-registered, reaching Cohen's $\kappa = 0.786$ (27/30 exact agreement) after one registered annotation-guideline revision round.

Answerability under Schema Evolution in Text-to-SQL

2026

First author — full manuscript complete (v1.0); not yet submitted

- Built a 1,382-instance benchmark over 20 Spider databases and seven schema-drift families, separating drift a model can observe from drift that is structurally invisible to it.
- Ran 27,384 generations across 39 experimental cells on free-tier Kaggle T4 GPUs, isolating where prompting interventions fail to help at all.
- Shipped a detector recovering 79.5% of the structurally invisible region at a 13.0% false-alarm rate, or 51.2% at zero false alarms, evaluated against 5,627 benign-update variants with labels double-annotated to $\kappa = 0.918$.

PROJECTS

Driftline — Fraud Modelling and Drift Monitoring

2026

Python, XGBoost, Redpanda/Kafka, PyFlink, Feast, MLflow, ONNX | [GitHub Link](#)

- Showed that evaluation design alone inflates measured performance by 0.2222 PR-AUC: an identical XGBoost configuration on 590K IEEE-CIS transactions scores 0.6974 on a random split against 0.4751 chronological.
- Measured forward PR-AUC decay from 0.4761 to 0.3680, then implemented PSI-triggered retraining with a shadow promotion gate — candidate 0.4802 against incumbent 0.2811 on the same shadow holdout.

Freight Rate ML — Leakage-Free Temporal Regression Benchmark

2026

Python, scikit-learn, gradient boosting, pandas | [GitHub Link](#)

- Built a reproducible temporal benchmark over 38,477 training rows and a 9,523-row temporal holdout, fitting coordinate lookups on training data only and recording library versions and a data hash so results replay exactly.
- Cut MAPE to 6.89% against an 8.28% linear baseline, and held boosted MAE at 154.21 versus 164.33 under an ablation removing market signals and coordinates, while reporting that the baseline retained better full-input MAE/RMSE.

EXPERIENCE

Software Developer Intern (Hybrid)

Oct 2024 – Mar 2025

Joget Inc. — enterprise workflow automation on the Joget DX low-code platform

Kuala Lumpur, Malaysia

- Shipped workflow automations and UI components used by client teams across [N] departments, owning the full SDLC in Agile sprints alongside senior engineers, QA and product managers.
- Removed manual re-entry steps from recurring client workflows by integrating REST APIs with external services.
- Maintained and released modules through CI/CD across [N] recurring production release cycles for enterprise clients.

AWARDS & LEADERSHIP

- Gold Medal and Best Video Award**, myHCI-UX Student Design Challenge (Oct 2025) — national inter-university UX design competition.
- Secretary Treasurer, SOFEA Society (2022–2023); organised AI workshops for 200+ students.